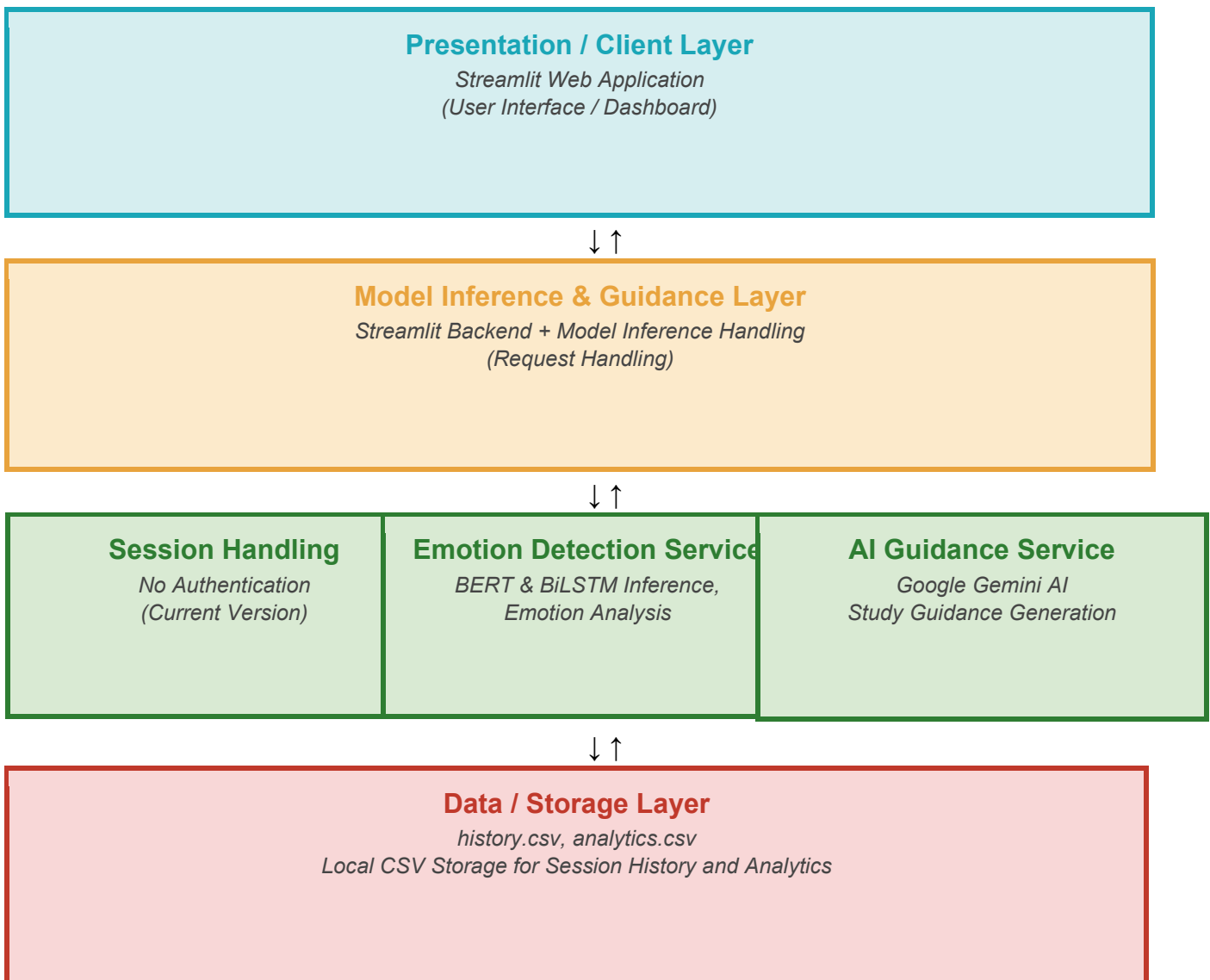


Solution Architecture

Date	28 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the interface for subject selection, learning problem input, viewing emotion analysis and model comparison, and downloading AI reports	Streamlit, HTML, CSS
Model Inference Layer	Runs the BERT and BiLSTM models to detect primary and mixed learning emotions with confidence scores	Python, TensorFlow, Keras, Hugging Face Transformers, PyTorch
AI Guidance Service	Generates personalized study guidance, explanations, and study plans based on the detected emotions	Google Generative AI SDK (Gemini 2.5 Flash)
Database / Storage	Stores session history, emotion results, and analytics data for future access and dashboard visualization	CSV Files (history.csv, analytics.csv)
Analytics & Visualization	Aggregates emotion trends and generates interactive charts for the analytics dashboard	Pandas, NumPy, Plotly
Testing Layer	Validates model predictions, application flow, and deployment readiness	Manual & automated testing (Python)